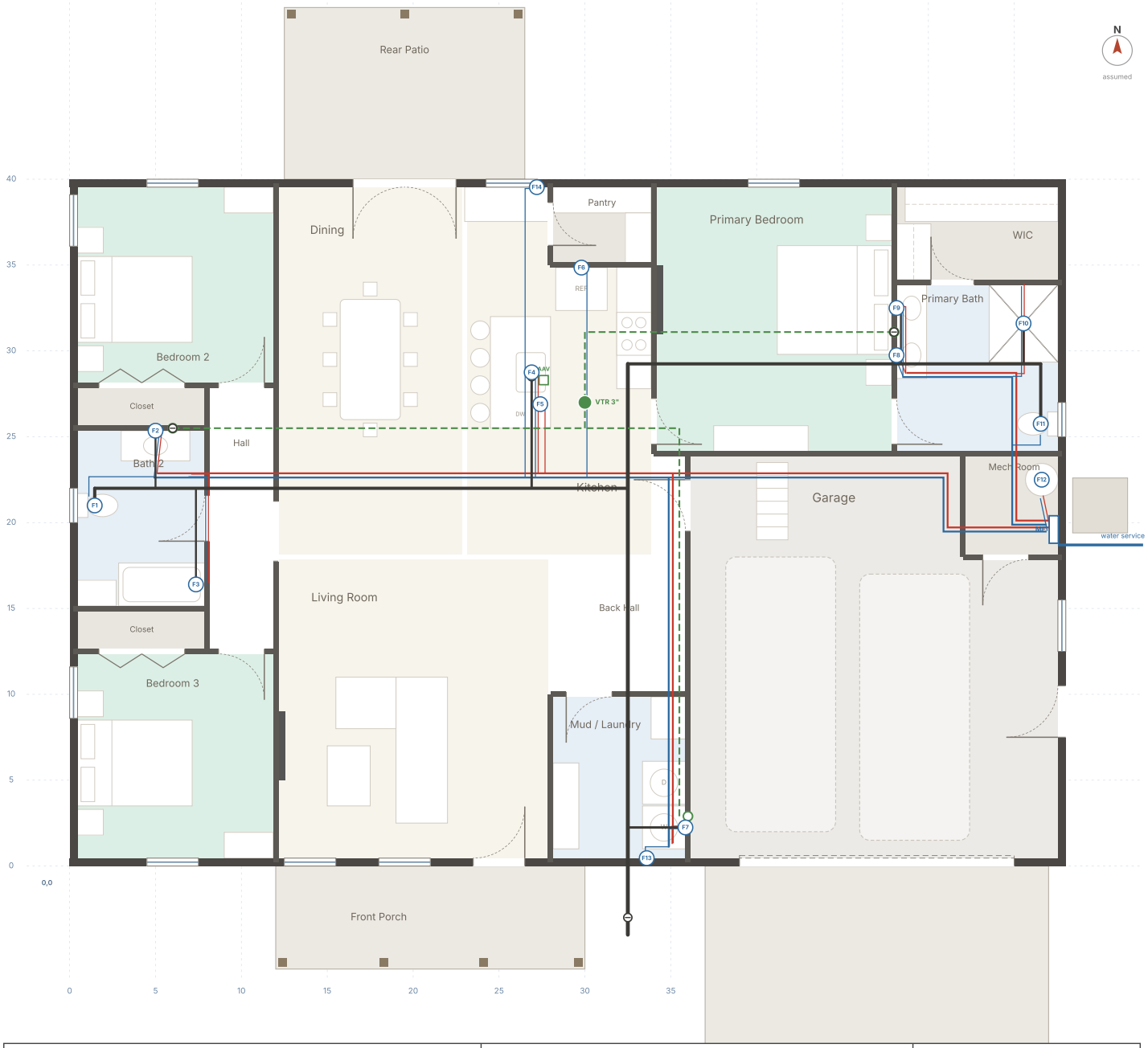


Waring Build BuildWise plan 3933 · 3 bed, 2 bath · 1,800 sq ft + 528 sq ft garage · 58'-0" x 40'-0" framing Slab on grade · 9' ceilings · 6:12 gable · grid 5', 0,0 at the south-west corner	**Plumbing plan** Drains and vents (DWV), PEX supply home runs, fixtures Schematic for pricing and coordination, not a stamped construction document. Verify on site.	**P-1** rev drawn for K. Christianson

P-1 Plumbing

drains under the slab (thicker = larger) vents: rise in the wall, run in the attic (dashed) cold PEX hot PEX F# fixture, CO# cleanout, MF manifold

Notes

1. Code basis: IRC 2021 chapters 25–33 (confirm the adopted edition). Drains PVC DWV schedule 40 under the slab at 1/4"/ft; the building drain is 4" (19 DFU, well under the 3" limit, 4" for a septic connection).
2. Every drain run is under the slab, laid before the pour, tested with a 10' head (P2503.5). Fixture drain positions are where the trap arm meets the wall or the floor.
3. Vents: each bathroom group is wet-vented through its lavatory (P3108), with one 2" dry vent rising in an interior wall. The three vents join in the attic and leave through one 3" vent on the rear roof slope, out of sight from the street.
4. Supply: PEX-A home runs from the manifold MF in the garage mech room, 1/2" to each fixture (3/4" to the tub and washer), under the slab in sleeves with no fittings below the slab. The bundle crosses under the garage slab: insulate it there, and keep every line on the warm side of the insulation in walls 13 and 14.
5. No supply lines in exterior walls: the toilets are fed up through the slab and the tub valve is on wall 8 (interior).
6. Water heater: 50 gal electric in the mech room on a pan with a drain, thermal expansion tank, T&P discharge to within 6" of the floor. A heat pump water heater fits the same space if the room is big enough for its air.

Open items

The particulars to settle on this sheet.

- Sewer or septic, and which side of the lot? The building drain is drawn leaving the front wall under the laundry.
- Water source: city or well? A well adds a pressure tank in the mech room (about 3' x 3' beside the water heater).
- Island sink vent: an AAV is drawn. Some jurisdictions want an island loop vent instead; confirm with the inspector.
- Water softener or filter at the service entry?
- Tub in Bath 2 (drawn) or a walk-in shower?

Fixture schedule

19 DFU in all (IRC Table P3004.1).

Tag	Fixture	Trap	DFU	Vent	Supply	Rough-in	Notes
F1	Bath 2 toilet	3"	3	V1	cold	8"	1.28 gpf, 12" rough from the west wall; supply up through the slab, not in the exterior wall
F2	Bath 2 lavatory	1-1/4"	1	V1	hot + cold	20"	trap arm into wall 9, vent V1 rises behind it
F3	Bath 2 tub / shower	1-1/2"	2	V1	hot + cold	48"	drain at the east end; valve and shower riser on wall 8 (hall side access panel optional)
F4	Kitchen sink (island)	1-1/2"	2	AAV	hot + cold	20"	double bowl with disposal; air admittance valve in the sink base (P3114), or an island loop vent if the AHJ requires
F5	Dishwasher	via F4	2	AAV	hot	6"	high loop to the disposal
F6	Refrigerator ice maker	—	0	—	cold	10"	recessed ice-maker box in wall 19, 1/4-turn valve
F7	Washer	2"	2	V3	hot + cold	42"	recessed washer box in wall 13 with a 2" standpipe; wall 13 faces the unheated garage: PEX on the warm side of the insulation
F8	Primary lavatory, south	1-1/4"	1	V2	hot + cold	20"	trap arm into wall 16

Tag	Fixture	Trap	DFU	Vent	Supply	Rough-in	Notes
F9	Primary lavatory, north	1-1/4"	1	V2	hot + cold	20"	trap arm into wall 16
F10	Primary shower	2"	2	V2	hot + cold	48"	centre drain, PVC liner or a foam pan; valve on wall 17 at 48", head at 80"
F11	Primary toilet	3"	3	V2	cold	8"	1.28 gpf, 12" rough from the east wall; supply up through the slab
F12	Water heater	—	0	—	hot + cold	—	50 gal electric (or heat pump water heater), thermal expansion tank, T&P discharge to within 6" of the garage floor, in a drain pan
F13	Hose bib, front	—	0	—	cold	24"	frost-free sillcock through wall 1 from the laundry, vacuum breaker
F14	Hose bib, rear	—	0	—	cold	24"	frost-free sillcock through wall 3 from the rear counter base, vacuum breaker

Drain schedule

Run	Size	Length	Slope	Notes
BD	4"	33'-3"	1/4"/ft (1/8" min)	building drain, 1/4"/ft, out under the laundry and the front wall to the sewer / septic; cleanout CO1 outside
B1	3"	31'-1"	1/4"/ft	Bath 2 branch, 1/4"/ft east under the hall and the great room to the building drain
B1a	3"	1'-0"	1/4"/ft	F1 closet bend
B1b	1-1/2"	3'-4"	1/4"/ft	F2 lav arm
B1c	1-1/2"	5'-7"	1/4"/ft	F3 tub
B2	2"	6'-9"	1/4"/ft	kitchen island: F4 and F5
B3	3"	24'-1"	1/4"/ft	primary bath branch, west under the primary bedroom and the kitchen
B3a	3"	3'-6"	1/4"/ft	F11 closet bend
B3b	2"	2'-4"	1/4"/ft	F10 shower
B3c	1-1/2"	3'-3"	1/4"/ft	F8 and F9 lavs
B4	2"	3'-0"	1/4"/ft	F7 washer standpipe

Vent schedule

Vent	Size	Rises in	Length	Notes
V1	2"	W9	34'-6"	Bath 2 group: F1, F2, F3 wet-vented through the lav; dry vent up in wall 9
V2	2"	W16	31'-1"	primary bath group: F8 to F11 wet-vented; dry vent up in wall 16
V3	2"	W13	37'-1"	washer standpipe vent up in wall 13
VTR	3"	roof, rear slope	—	one 3" vent through the roof on the rear slope, 6" above the roof, flashed boot (IRC P3103)
AAV	1-1/2"	island sink base	—	air admittance valve in the island sink base, 4" above the trap arm, accessible (P3114)

Supply runs

Lengths are along the drawn route plus 4' for the rise; order by the coil.

Run	What	Length	Notes
TW	bundle	55'-2"	west home-run bundle under the garage slab and the house, in sleeves through the garage foundation; insulate under the garage slab
TP	bundle	17'-0"	primary bath bundle, north under wall 14

Run	What	Length	Notes
TS	bundle	21'-7"	laundry and front hose bib bundle
HF1	1/2" to F1 Bath 2 toilet	8'-11"	cold only
HF2	1/2" to F2 Bath 2 lavatory	6'-11"	hot and cold
HF3	3/4" to F3 Bath 2 tub / shower	11'-4"	hot and cold
HF4	1/2" to F4 Kitchen sink (island)	10'-0"	hot and cold
HF5	1/2" to F5 Dishwasher	8'-2"	hot only
HF6	1/2" to F6 Refrigerator ice maker	16'-1"	cold only
HF14	1/2" to F14 Hose bib, rear	21'-5"	cold only
HF7	3/4" to F7 Washer	5'-11"	hot and cold
HF13	1/2" to F13 Hose bib, front	6'-2"	cold only
HF8	1/2" to F8 Primary lavatory, south	5'-7"	hot and cold
HF9	1/2" to F9 Primary lavatory, north	8'-4"	hot and cold
HF10	1/2" to F10 Primary shower	9'-9"	hot and cold
HF11	1/2" to F11 Primary toilet	6'-2"	cold only
HF12	3/4" to F12 Water heater	6'-4"	hot and cold

Cleanouts

Tag	Where
CO1	two-way cleanout outside the front wall, at grade
CO2	base of V1 in wall 9, 12" AFF, access behind the vanity
CO3	base of V2 in wall 16, access in the vanity
CO4	island sink trap, in the sink base
CO5	washer standpipe

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The plumber takes off the final list.

Item	Qty	Notes
PVC DWV 1-1/2" sch 40	2 × 10'	13' measured +15%
PVC DWV 2" sch 40	2 × 10'	13' measured +15%
PVC DWV 2" vent sch 40	12 × 10'	103' measured +15%
PVC DWV 3" sch 40	7 × 10'	60' measured +15%
PVC DWV 4" sch 40	4 × 10'	34' measured +15%
PEX-A 1/2" (red and blue)	5 × 100' coils	about 356' of home runs
PEX-A 3/4" and 1"	by the run	tub, washer and water heater runs, service
PEX manifold, 12-port hot / 12-port cold, with valves	1	MF in the mech room
Water heater, 50 gal electric (or heat pump), pan, expansion tank	1	
Sleeves for PEX through the slab and the garage foundation	18	protect every run where it leaves the slab

Item	Qty	Notes
Frost-free sillcocks, 12"	2	F13, F14
Washer box with valves and 2" drain	1	F7